Task 1:

```
.386
.model flat, stdcall
.stack 4096
ExitProcess proto, dwExitCode:dword
include Irvine32.inc

.data
filename BYTE "MyFile.txt", 0
fileHandle HANDLE ?

buffer BYTE 1024 DUP(0)
bytesRead DWORD ?

; Messages
msg1 BYTE "File content: ", 0dh, 0ah, 0
msg2 BYTE "Message to write: ", 0dh, 0ah, 0
msg3 BYTE "Bytes written: ", 0
.code
main PROC
        ; Create file
        mov edx, OFFSET filename
        call CreateOutputFile
        mov fileHandle, eax

        ; Check if file creation was successful
        cmp eax, INVALID_HANDLE_VALUE
        je exit_program

        ; Write to File
        mov edx, OFFSET msg2
        call WriteString

        ; Get input from user
        mov edx, OFFSET buffer
        mov ecx, SIZEOF buffer - 1
        call ReadString

        ; Write to file
        mov eax, fileHandle
        mov edx, OFFSET buffer
        mov ecx, eax            ; ECX should contain the count of bytes to write
        call StrLength          ; Get length of string in EAX
        mov ecx, eax            ; Move length to ECX
```

```
        mov eax, fileHandle
        call WriteToFile

        ; Display bytes written
        mov edx, OFFSET msg3
        call WriteString
        call WriteDec
        call Crlf

        ; Close file
        mov eax, fileHandle
        call CloseFile

        ; Read file content
        mov edx, OFFSET msg1
        call WriteString

        mov edx, OFFSET filename
        call OpenInputFile
        mov fileHandle, eax

        ; Check if file opening was successful
        cmp eax, INVALID_HANDLE_VALUE
        je exit_program

        ; Read from file
        mov edx, OFFSET buffer
        mov ecx, SIZEOF buffer - 1
        call ReadFromFile
        mov bytesRead, eax

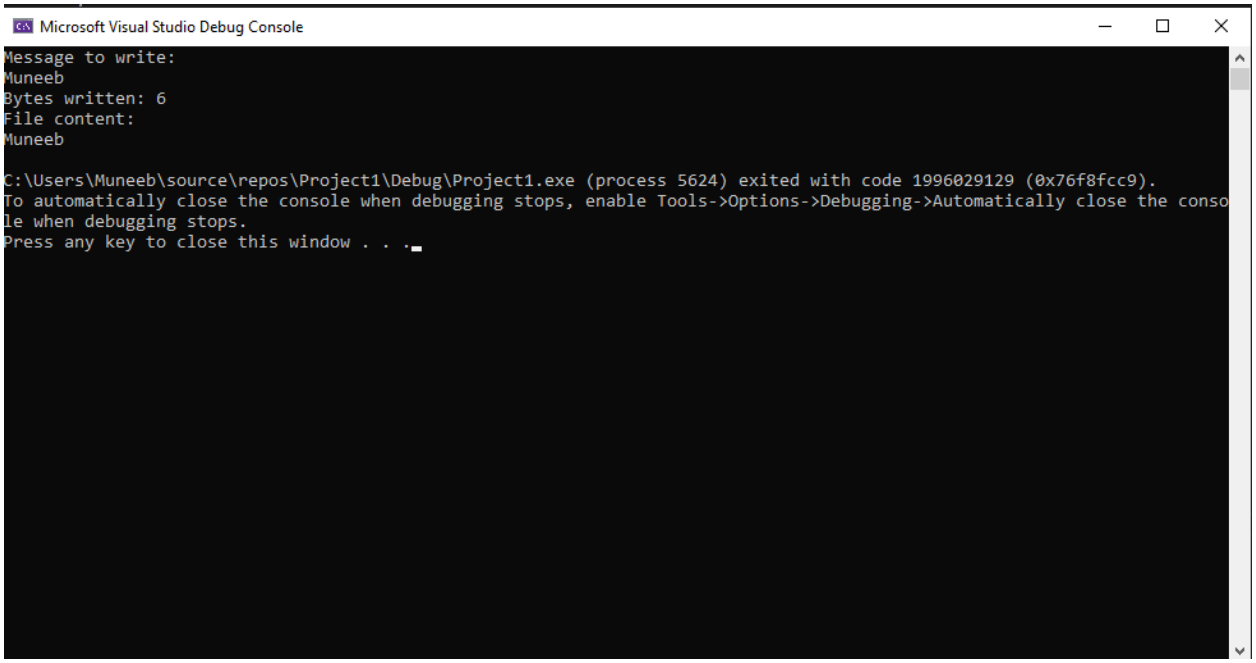
        ; Display file content
        mov edx, OFFSET buffer
        call WriteString
        call Crlf

        ; Close file
        mov eax, fileHandle
        call CloseFile

        exit_program:

        call ExitProcess
main ENDP
```

END main



Task 2:

```
INCLUDE Irvine32.inc
.386
.model flat, stdcall
.stack 4096

.data
fileName        db "MyFile.txt", 0
fileHandle  dword ?
msgPrompt   db "Enter a character: ", 0
charInput   db ?
newline         db 13,10,0

.code
main PROC

        ; Create the file
        mov edx, OFFSET fileName
        call CreateOutputFile
        mov fileHandle, eax

        mov ecx, 5

InputLoop:
```

```
        ; Prompt user
        mov edx, OFFSET msgPrompt
        call WriteString

        ; Read a single character
        call ReadChar
        mov charInput, al


        mov al, charInput
        call WriteChar
        call Crlf

        ; Write character to file
        mov eax, fileHandle
        mov edx, OFFSET charInput
        push ecx
        mov ecx, 1              ; write 1 byte
        call WriteToFile
        pop ecx

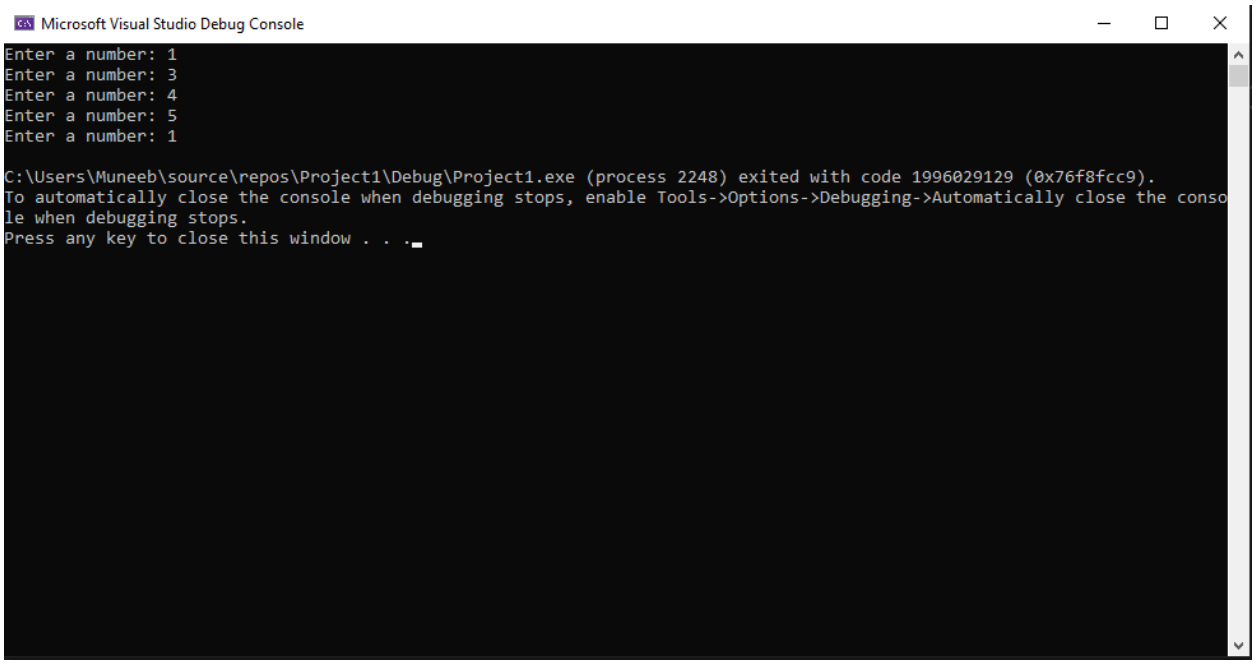
        loop InputLoop


        mov eax, fileHandle
        mov edx, OFFSET newline
        mov ecx, LENGTHOF newline - 1
        call WriteToFile

        ; Close file
        mov eax, fileHandle
        call CloseFile

        call ExitProcess
main ENDP

END main
```



Task 3:

```
INCLUDE Irvine32.inc
.386
.model flat, stdcall
.stack 4096

.data
fileName        db "MyFile.txt", 0      ; File to open
fileHandle  dword ?             ; Handle to the file
buffer   db 300 dup(0)          ; Small buffer (300 bytes)
bytesRead   dword ?
msgHeader   db "File contents:",13,10,0

.code
main PROC


        mov edx, OFFSET msgHeader
        call WriteString


        mov edx, OFFSET fileName
        call OpenInputFile
        mov fileHandle, eax

ReadLoop:
        ;Read up to 300 bytes into buffer
```

```
        mov eax, fileHandle
        mov edx, OFFSET buffer
        mov ecx, 300
        call ReadFromFile
        mov bytesRead, eax

        ;EOF
        cmp eax, 0
        je DoneReading

        ;Write buffer to console
        mov edx, OFFSET buffer
        call WriteString

        jmp ReadLoop

DoneReading:
        ;Close file
        mov eax, fileHandle
        call CloseFile

        ;Exit program
        call ExitProcess
main ENDP

END main
```

